

Homework 4

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Problem 1

Given: $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, σ^2 known.

Goal: CRLB lower bound for $\text{Var}(\hat{\mu})$ among unbiased estimators

Plan

- (i) Write $\ell(\mu)$ CRLB needs Fisher information from log-likelihood curvature
- (ii) compute $\ell''(\mu)$
- (iii) compute $I(\mu) = -E[\ell''(\mu)]$
- (iv) Apply CRLB: $\text{Var}(\hat{\mu}) \geq 1/I(\mu)$

① Log-likelihood take logs to turn products into sums and make easy derivatives

$$\ell(\mu) = \sum_{i=1}^n \log f(x_i | \mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

② First derivative differentiate to get the score.

$$\frac{d}{d\mu} (x_i - \mu)^2 = 2(x_i - \mu)(-1) = -2(x_i - \mu)$$

$$\ell'(\mu) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (-2)(x_i - \mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

③ Second derivative curvature gives information

$$\ell''(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n \frac{d}{d\mu} (x_i - \mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (-1) = -\frac{n}{\sigma^2}$$

④ Fisher information

$$I(\mu) = -\mathbb{E}[l''(\mu)] = -\left(-\frac{n}{\sigma^2}\right) = \frac{n}{\sigma^2}$$

⑤ CRLB

$$\text{Var}(\hat{\mu}) \geq \frac{1}{I(\mu)} = \frac{1}{n/\sigma^2} = \frac{\sigma^2}{n}$$

Problem 2

Given: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$. Let $S = \sum_{i=1}^n X_i$

2.a) Derive the Fisher Information $I(p)$

Plan

① Write $l(p)$

② compute $l'(p)$, $l''(p)$

③ Take expectation and negate: $I(p) = -\mathbb{E}[l''(p)]$

① Likelihood: $L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^S (1-p)^{n-S}$

② Log-Likelihood: $l(p) = S \log p + (n-S) \log(1-p)$

③ 1st Derivative: $l'(p) = \frac{S}{p} - (n-S) \frac{1}{1-p}$

④ 2nd Derivative: $l''(p) = -\frac{S}{p^2} - \frac{n-S}{(1-p)^2}$

⑤ Information: use $\mathbb{E}[S] = np$, $\mathbb{E}[n-S] = n(1-p)$

$$I(p) = -\mathbb{E}[l''(p)] = \mathbb{E}\left[\frac{S}{p^2}\right] + \mathbb{E}\left[\frac{n-S}{(1-p)^2}\right] = \frac{np}{p^2} + \frac{n(1-p)}{(1-p)^2}$$

$$I(p) = \frac{n}{p} + \frac{n}{1-p} = \frac{n}{p(1-p)}$$

2.b) Show $I(p) \rightarrow \infty$ as $p \rightarrow 0$ or $p \rightarrow 1$

Plan check the denominator from the 2.a) result, $p(1-p)$ and see what happens at the edges.

$$I(p) = \frac{n}{p(1-p)} \begin{cases} \rightarrow p \rightarrow 0^+, p(1-p) \rightarrow 0^+ \Rightarrow I(p) \rightarrow \infty \\ \rightarrow p \rightarrow 1^-, 1-p \rightarrow 0^+ \Rightarrow p(1-p) \rightarrow 0^+ \Rightarrow I(p) \rightarrow \infty \end{cases}$$

Problem 3

Same normal model as Problem 1.

3.a) Plan write $\bar{X}$ as a scaled sum and use variance rules + Independence.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \rightarrow \text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right)$$

Independence gives $\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2$. Hence

$$\text{Var}(\bar{X}) = \frac{1}{n^2} (n\sigma^2) \rightarrow \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

3.b) Compare to CRLB

From Problem 1. CRLB is $\sigma^2/n \rightarrow$ We found $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$. So $\bar{X}$ meet the bounds

3.c) Conclude whether $\bar{X}$ is efficient.

If an unbiased estimator reach CRLB, it's efficient.

$\bar{X}$ is unbiased for μ and reach CRLB, Then is efficient

Problem 4

Bernoulli model, estimator $T = X_1$ for p .

a) Is T unbiased? Justify your answer.

T is unbiased for p if $\mathbb{E}[T] = p$

Since $T = X_1$ and $X_1 \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$, we have $P(X_1 = 1) = p$ and $P(X_1 = 0) = 1 - p$

Therefore, using the definition of expectation for a discrete random variable

$$\mathbb{E}[T] = \mathbb{E}[X_1] = \sum_{x \in \{0,1\}} x P(X_1 = x) = 0 \cdot (1-p) + 1 \cdot p = p$$

Hence $\mathbb{E}[T] = p$, so T is an unbiased estimator of p .

b) Compute $V(T)$

For Bernoulli(p). $\rightarrow \text{Var}(T) = \text{Var}(X_1) = p(1-p)$

4.c) Compare the variance in part b) to the CRLB and comment on efficiency.

$$\text{Var}(\hat{p}) \geq \frac{1}{I(p)} = \frac{1}{\frac{n}{p(1-p)}} = \frac{p(1-p)}{n}$$

$$\text{but } \text{Var}(T) = p(1-p) = n \cdot \frac{p(1-p)}{n}$$

So T is not efficient (unless $n=1$)

$$\text{Efficiency ratio: } \text{eff}(T) = \frac{\text{CRLB}}{\text{Var}(T)} = \frac{p(1-p)/n}{p(1-p)} = \frac{1}{n}$$

Problem 5

Given: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exponential}(\lambda)$, $f(x|\lambda) = \lambda e^{-\lambda x}$, $x \geq 0$

a) Determine $I(\lambda)$

(i) Likelihood:
$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

(ii) log-likelihood

$$l(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

(iii) 1st Der

$$l'(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n x_i$$

(iv) 2nd Der

$$l''(\lambda) = -\frac{n}{\lambda^2}$$

(v) Fisher Information

$$I(\lambda) = -E[l''(\lambda)] = -\left(-\frac{n}{\lambda^2}\right) = \frac{n}{\lambda^2}$$

b) Let $\theta = \log(\lambda)$. Compute $I(\theta)$. Lets rewrite l in θ using $\lambda = e^\theta$, then

compute $-E[l''(\theta)]$

$$\lambda = e^\theta, \quad l(\theta) = n \log(e^\theta) - e^\theta \sum_{i=1}^n x_i = n\theta - e^\theta \sum_{i=1}^n x_i$$

$$l'(\theta) = n - e^\theta \sum_{i=1}^n x_i, \quad l''(\theta) = -e^\theta \sum_{i=1}^n x_i = -\lambda \sum_{i=1}^n x_i$$

Use $E[x_i] = \frac{1}{\lambda} \Rightarrow E\left[\sum_{i=1}^n x_i\right] = \frac{n}{\lambda}$:

$$I(\theta) = -E[l''(\theta)] = -\left(-\lambda \cdot \frac{n}{\lambda}\right) = n$$

c) compute $\frac{d\lambda}{d\theta}$: $\lambda = e^\theta \Rightarrow \frac{d\lambda}{d\theta} = \lambda$, now lets verify $I(\theta) = I(\lambda) \left(\frac{d\lambda}{d\theta}\right)^2$

$$I(\lambda) \left(\frac{d\lambda}{d\theta}\right)^2 = \frac{n}{\lambda^2} \cdot \lambda^2 = n = I(\theta)$$

Problem 6

Example where CRLB exists but no estimator reaches it.

Take $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with μ and σ^2 unknown, and target σ^2

(i) CRLB exists?

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

Diff w.r.t σ^2

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (X_i - \mu)^2$$

$$\text{2nd Der. } \frac{\partial^2 \ell}{\partial (\sigma^2)^2} = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^n (X_i - \mu)^2$$

expectation using $\mathbb{E}[\sum (X_i - \mu)^2] = n\sigma^2$:

$$I(\sigma^2) = -\mathbb{E}\left[\frac{\partial^2 \ell}{\partial (\sigma^2)^2}\right] = -\left(\frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \cdot n\sigma^2\right) = \frac{n}{2\sigma^4} \quad \text{so CRLB} \rightarrow$$

$$\text{Var}(\hat{\sigma}^2) \geq \frac{1}{I(\sigma^2)} = \frac{2\sigma^4}{n}$$

(ii) Unbiased estimator and its variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \mathbb{E}[S^2] = \sigma^2$$

Know fact for normal samples: $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$
 [Reference from: <https://math.stackexchange.com/questions/47009>]

Variance of chi-square: $\text{Var}(\chi_k^2) = 2k \rightarrow$

$$\text{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1) \rightarrow \text{"Scale back"} \rightarrow S^2 = \frac{\sigma^2}{n-1} Y$$

$$(Y \sim \chi_{n-1}^2) \Rightarrow \text{Var}(S^2) = \left(\frac{\sigma^2}{n-1}\right)^2 \text{Var}(Y) = \frac{\sigma^4}{(n-1)^2} \cdot 2(n-1) = \frac{2\sigma^4}{n-1}$$

(ii) Compare

$$\frac{2\sigma^4}{n-1} > \frac{2\sigma^4}{n}$$

So the unbiased estimator S^2 does not attain CRLB; and in this model the equality condition required to attain the bound cannot be satisfied for σ^2 (with μ unknown), so the CRLB is not attainable.